Survey analysis

The labs and practice sets of Day 3. You type your work in the do-file named on each page; this booklet is the printed copy to read, plan and write notes on. Core first, then Extension, then Challenge.

How the checks work. Every task in the do-file ends with a CHECK line. Run the file with Ctrl + D: a silent CHECK means your answer is correct; red text under a CHECK means look at that task again. Never edit a CHECK line. The solutions booklet explains the reasoning behind each answer; read it after you have tried.

Lab 1: Document the design

File `C:/NBS_Stata/day3/lab_D3_1.do`
inputs data/lfs_teach.dta
outputs logs/lab_D3_1.log, output/design_D3.xlsx

The situation
The methodology unit asks for a one-page note on the design of the Q1 2026 LFS: strata, PSUs, weights and the svyset line, with every number produced by Stata, not copied from a report.

Task 1 Settings and the diary — Core

Write the five top lines of every course do-file. The diary is logs/lab_D3_1.log.

Task 2 The weights — Core

Open lfs_teach.dta. Summarize wgt_final and display the sum of the weights with a thousands comma.

Task 3 PSUs per district — Core

Create first_psu (1 on the first row of each PSU) and n_psu (the number of PSUs in the person's district). Tabulate n_psu, one row per PSU.

Task 4 Rebuild the final weight — Core

Create check_w = base weight times district adjustment, and write one assert that proves it equals wgt_final (allow for rounding).

Task 5 Declare and describe the design — Core

Declare the design with svyset and run svydescribe. Then complete the five-line note below from the output.

Task 6 PSUs per district into Excel — Extension

One row per district with its number of PSUs, exported to output/design_D3.xlsx. Hint: table … if first_psu, then collect export.

Task 7 Districts interviewed in one month only — Challenge

Count, for each district, the months in which it was interviewed (n_month). How many districts were interviewed in one month only, and how does their mean weight compare with the others?

A correct lab. the whole file runs from top to bottom with no red text, ends with "end of do-file", logs/lab_D3_1.log exists, and the five-line design note in Task 5 is filled in.

Lab 2: The estimates release

File `C:/NBS_Stata/day3/lab_D3_2.do`
inputs data/lfs_teach.dta
outputs logs/lab_D3_2.log, output/release_q1.xlsx

The situation
The labour statistics section prepares the Q1 2026 release: unemployment and participation rates by sex and by age group, with standard errors and intervals, in one Excel file, plus a notes sheet that documents the design and flags imprecise figures.

Task 1 Settings, data, design — Core

The five top lines (diary logs/lab_D3_2.log), open lfs_teach.dta and declare the design.

Task 2 Four flags — Core

wap (15 and over), lf (employed or unemployed), unemp, emp: 0/1, never missing.

Task 3 Unemployment by sex and by age group — Core

Estimate the unemployment rate by sex and copy r(table)' into S; then by age group into A. Run estat cv after the age groups.

Task 4 Participation by sex and by age group — Core

Same for the participation rate: matrices PS (sex) and PA (age).

Task 5 The release file — Core

Keep rate, SE, lower, upper; multiply by 100; give readable row and column names. Write S and A to the sheet Unemployment, PS and PA to the sheet Participation, in output/release_q1.xlsx.

Task 6 Notes and flags — Extension

A sheet Notes: a title, the svyset line, the source, and one line naming every estimate with a CV above 30% (placeholder until NBS confirms its threshold).

Task 7 Time-related underemployment — Challenge

The share of the employed who are time-related underemployed (ilo_joball_tru), with its CV. Look at how ilo_joball_tru is coded before you use it. Call your 0/1 flag tru.

A correct lab. the whole file runs with no red text, ends with "end of do-file", and output/release_q1.xlsx has the sheets Unemployment, Participation and Notes.

Practice sets

File `C:/NBS_Stata/day3/practice_D3.do`. One block per part of the day; each opens its own data, so the parts can be done in any order. Type your number after `nbscheck` where the file asks for it.

Part 1: Weights

P1.1 Core Weighted count of men and women with [iweight = wgt_final]. Type the weighted number of women, rounded.

P1.2 Core Sum of the weights by region with tabstat. Type the sum for Perseverance, rounded.

P1.3 Core Weighted share of women in percent, one decimal.

P1.4 Extension The largest single weight in the file, one decimal.

P1.5 Challenge How many persons have a weight above 100? List their district, PSU and household.

Part 2: The design

P2.1 Core How many PSUs are in the file? Tag the first row of each PSU with egen tag(), then count the tags.

P2.2 Core The largest number of PSUs in one district.

P2.3 Core Prove wgt_final = wgt_base * wgt_adj with one assert. Your assert is the check.

P2.4 Extension Median number of households per PSU.

P2.5 Challenge Prove with one line that every household lies in one PSU only.

Part 3: Declaring the design

P3.1 Core Declare the design and run svydescribe. Type the number of strata.

P3.2 Core The smallest number of observations in a PSU.

P3.3 Core Save the declared file as output/p3_svy.dta, open it again and show the design with svyset.

P3.4 Extension svydescribe ilo_job1_how_usual: how many persons have complete data?

P3.5 Challenge Declare the design without strata and run svydescribe. How many strata does Stata see? Then restore the correct declaration.

Part 4: Estimates (design and four flags ready)

P4.1 Core Design-based mean age, two decimals.

P4.2 Core Number of employed persons: type the lower bound of the 95% interval, rounded.

P4.3 Core Number of unemployed persons, rounded.

P4.4 Extension Employment-to-population ratio in percent, one decimal.

P4.5 Challenge Design effect of the participation rate, two decimals.

Part 5: Subpopulations

P5.1 Core Participation rate of women in percent, one decimal.

P5.2 Core The highest regional unemployment rate, percent, one decimal.

P5.3 Core Youth (15-24) unemployment rate, percent, one decimal.

P5.4 Extension Unemployment rate of young women, percent, one decimal.

P5.5 Challenge Employment-to-population ratio by region: the lowest, percent, one decimal.

Part 6: Survey tables

P6.1 Core Weighted percentage of women among the 15+, one decimal.

P6.2 Core Percentage of the population in Perseverance, one decimal.

P6.3 Core Weighted row percentages of status by region: the unemployed percentage in West.

P6.4 Extension Column percentages: the share of the unemployed who live in North, one decimal.

P6.5 Challenge Weighted count of people in Central 2, rounded.

Part 7: Precision

P7.1 Core CV of the unemployment rate in Central 2, one decimal.

P7.2 Core CV of the participation rate in West, one decimal.

P7.3 Core Interviews behind the Perseverance unemployment rate (estat size after the regional rates).

P7.4 Extension SE of the national unemployment rate without strata, four decimals. Restore the design afterwards.

P7.5 Challenge Time-related underemployment among the employed: build tru = (ilo_joball_tru == 1); rate in percent, one decimal.

Part 8: The estimates table (close Excel first)

P8.1 Core After the regional unemployment rates, list r(table). How many rows does it have?

P8.2 Core Copy and transpose it into T. Type the SE of West, four decimals.

P8.3 Core Write T to output/p8_estimates.xlsx, with names.

P8.4 Extension Keep rate, SE, lower and upper only, in matrix R.

P8.5 Challenge Participation by region in percent (rate, lower, upper) with region names. Type North's upper bound.